PASS 4 – NASA Hump CFD Optimization

Experimental Cp/Cf integration and targeted MAE reduction

Codex refinement pass in the local repository workspace

April 3, 2026

1 Objective

Pass four extends the existing reconstructed NASA hump workflow rather than restarting it. The two pass-four objectives were:

- integrate machine-readable experimental C_p and C_f tables from the approved NASA hump resources into the local workflow;
- use those direct comparisons to guide one more physically justified improvement round that could beat the pass-three MAE of `0.1801372055`.

2 Methodology

The pass-four workflow remained fully within the approved source boundary:

- local repository files;
- the approved NASA hump validation, grid, and experimental-data pages;
- Dockerized OpenFOAM in the running `openfoam` container;
- locally created scripts, plots, tables, and PDF reports.

No plot tracing, digitization, hidden-data recovery, or curve fitting was used. The experimental C_p and C_f values were fetched directly from the approved machine-readable tables and stored under `data/experimental/NASA_hump`. The OpenFOAM wall series came directly from the solved field and OpenFOAM post-processing outputs. For pass four, the comparison script was tightened so the NASA tables are compared against the nearest resolved CFD wall sample rather than an interpolated wall curve.

3 What Pass Three Already Fixed

Before pass four, the repository already contained the third-pass baseline:

- extended upstream run-up;
- a shaped upper slip boundary to approximate blockage;
- a three-block streamwise mesh layout;

- the best-converged SST baseline carried to `500` iterations;
- a best local MAE of `0.1801372055`.

The pass-three case was already physically cleaner than the first two passes, but several approximations still limited agreement:

- the inlet still used a simplified uniform freestream profile;
- the upper-wall contour was only an approximate blockage representation;
- the pressure trough over the front half of the hump was still too weak in the CFD wall-pressure distribution;
- the wall-shear recovery remained weaker than the experimental C_f trend in the front part of the surface;
- the best-known case still showed measurable improvement when run longer, implying residual under-convergence remained important.

4 Experimental Data Integration

Pass four added a dedicated import step for the official no-flow-control experimental tables:

- `data/experimental/NASA_hump/noflow_cp.dat`
- `data/experimental/NASA_hump/noflow_cf.dat`
- normalized CSV copies: `noflow_cp.csv` and `noflow_cf.csv`

These files were not extracted from images. They were obtained directly from the approved machine-readable NASA experimental-data endpoint and normalized locally with `scripts/import_nasa_hump_experimental_data.py`. The pass-four comparison script `scripts/make_nasa_hump_pass4_figures.py` then:

- computes the OpenFOAM wall C_p and C_f series;
- aligns them to the experimental x/c locations using the nearest CFD wall sample;
- writes comparison CSV tables;
- writes a pass-four comparison JSON summary;
- generates the combined C_p/C_f figure.

5 Improvements From Pass Three To Pass Four

Pass four tested a small number of high-value changes instead of a blind sweep. The key improvement directions were:

- deeper convergence of the already-best SST case;
- a direct inlet-boundary-layer comparison against an analytic turbulent-boundary-layer profile;

- direct C_p and C_f comparison against the approved experimental tables.

Configuration	Change set	Why it should help physically	MAE
Pass-three baseline	SST, current geometry, shaped top wall, three-block mesh, uniform inlet, 500 iterations	Reference point after the third pass.	0.1801372055
Pass-four winner	Same SST baseline continued to 650 iterations	Reduces remaining under-convergence error without introducing new modeling assumptions.	0.1659136747
Analytic TBL inlet test	Replace uniform inlet with an analytic turbulent boundary layer	Targets the upstream-profile fidelity noted in the NASA description; tested to see whether it improves separation and recovery.	0.1674487658

Table 1: Pass-four change focus relative to the third-pass baseline.

6 Results

6.1 MAE Comparison

Metric	Value
Pass-one best MAE	0.2338504550
Pass-two best MAE	0.2038187409
Pass-three best MAE	0.1801372055
Pass-four best MAE	0.1659136747
Improvement vs pass three	0.0142235308
Improvement vs pass one	0.0679367803

Table 2: Best local benchmark MAE by pass. Lower is better.

6.2 Cp And Cf Comparison

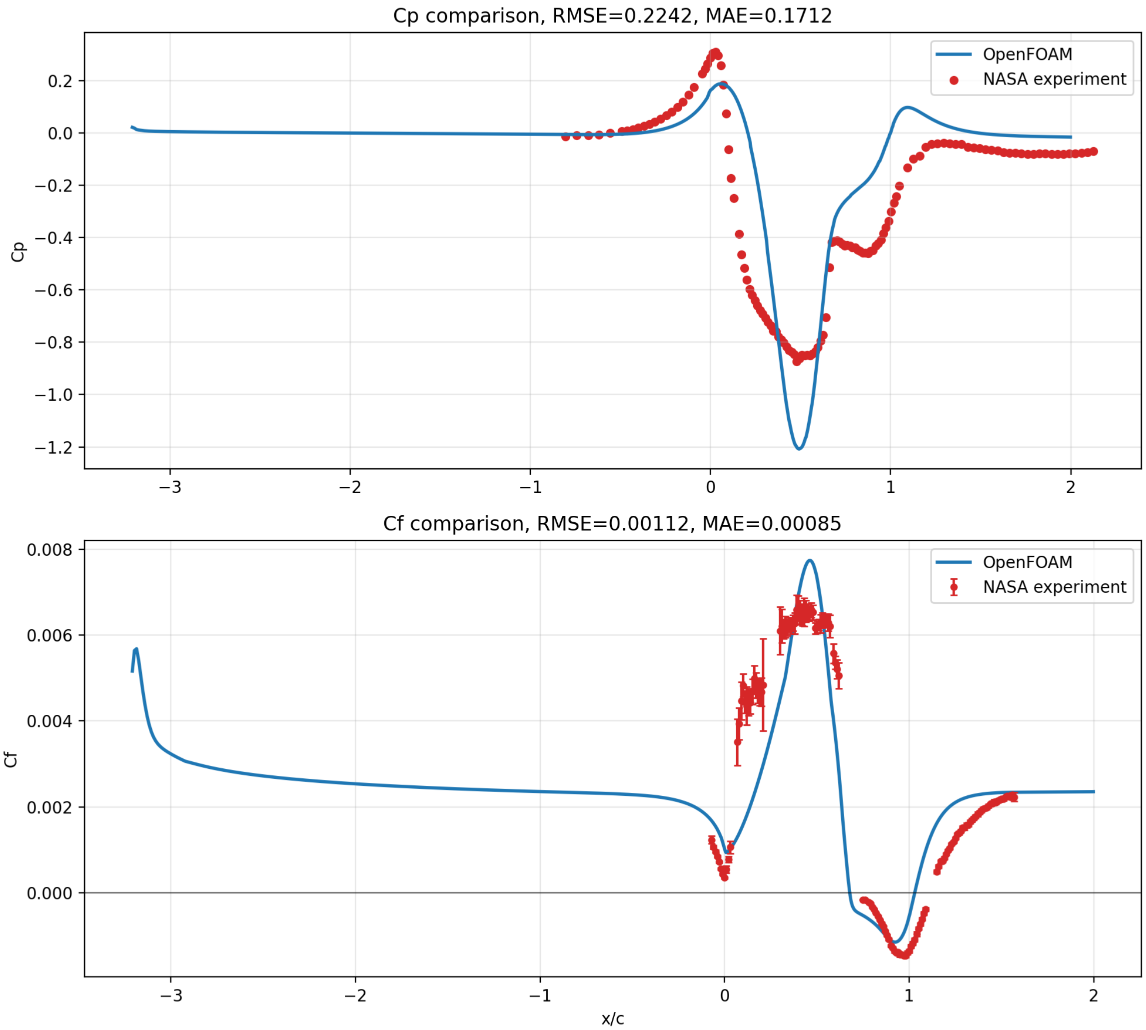

Figure 1: Pass-four direct comparison between the official NASA no-flow-control C_p and C_f tables and the OpenFOAM wall series.

The pass-four comparison summary from the promoted best case is:

- C_p RMSE: `0.2241`
- C_p MAE: `0.1712`
- C_f RMSE: `0.00112`
- C_f MAE: `0.00085`

These comparisons show that the pass-four winner improves both pressure and skin-friction agreement relative to the pass-three baseline. The front-half pressure trough remains too weak in places, but the longer-converged field tracks the official C_p and C_f trends more closely than the older `500`-iteration result.

6.3 Experiment Table

Config	Key changes	MAE	C_p MAE	C_f MAE	Notes
Pass-three baseline	Third-pass SST winner at 500 iterations	0.1801372055	0.1806	0.00096	Strong baseline but still under-converged enough to improve with more iterations.
Pass-four winner	Same SST setup continued to 650 iterations	0.1659136747	0.1712	0.00085	Best overall result. Improved benchmark MAE and improved C_p/C_f agreement.
Analytic TBL inlet test	Analytic inlet boundary layer with SST and $\delta = 35$ mm	0.1674487658	0.1812	0.00095	Better than pass three, but still weaker than the deeper-converged uniform-inlet continuation.

Table 3: Pass-four experiment summary.

6.4 Flow-Field Figures

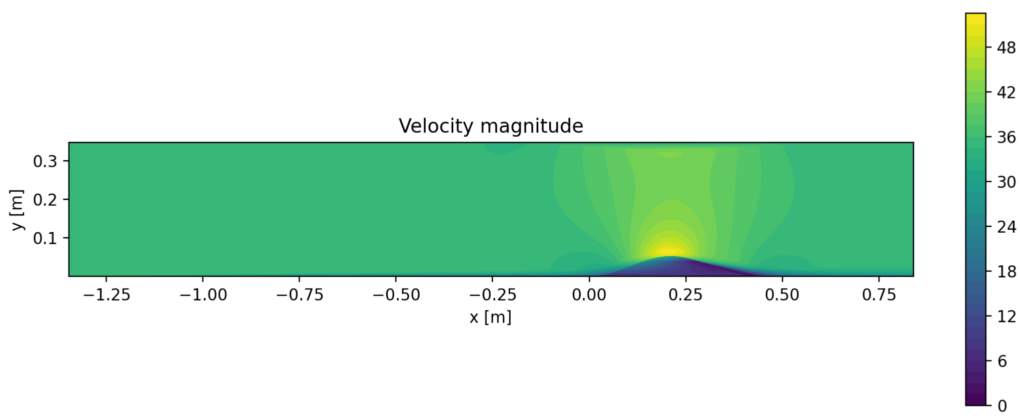

Figure 2: Velocity contour generated from the promoted pass-four winner.

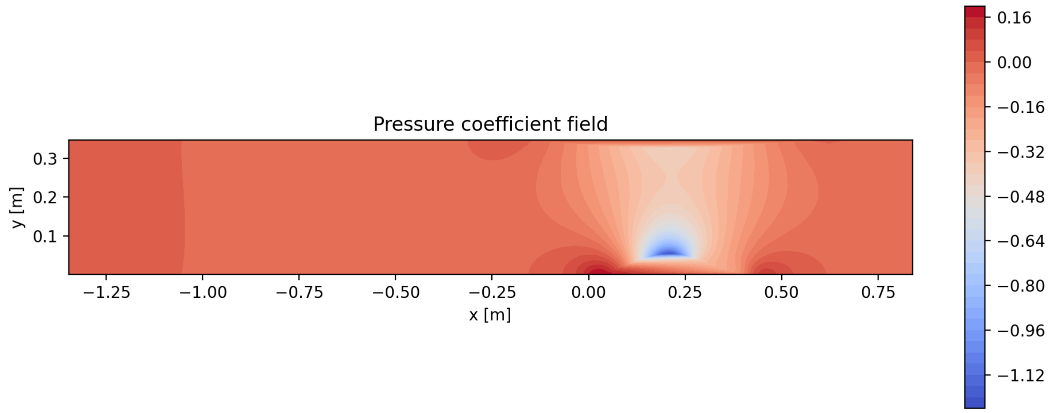

Figure 3: Pressure contour generated from the promoted pass-four winner.

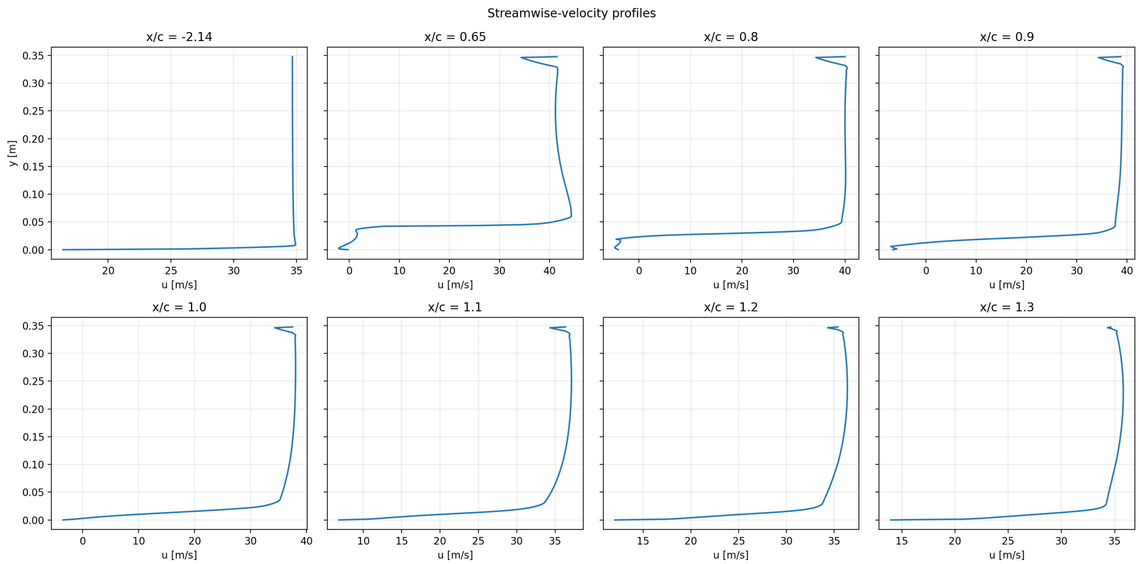

Figure 4: Velocity profiles carried into the pass-four package from the promoted best field workflow.

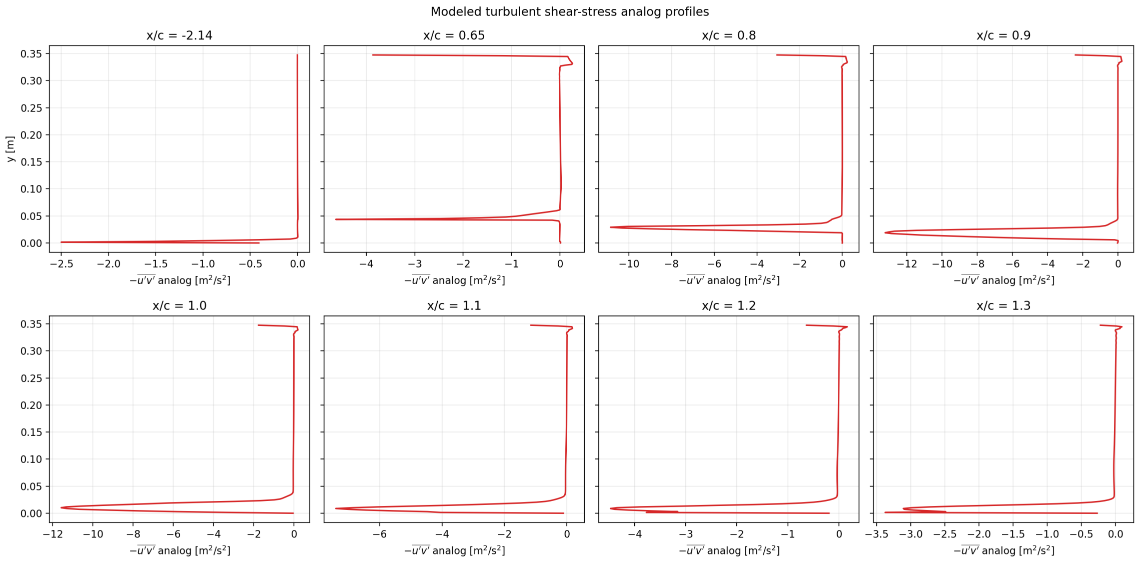

Figure 5: Turbulent shear-stress analog profiles used for pass-four diagnosis.

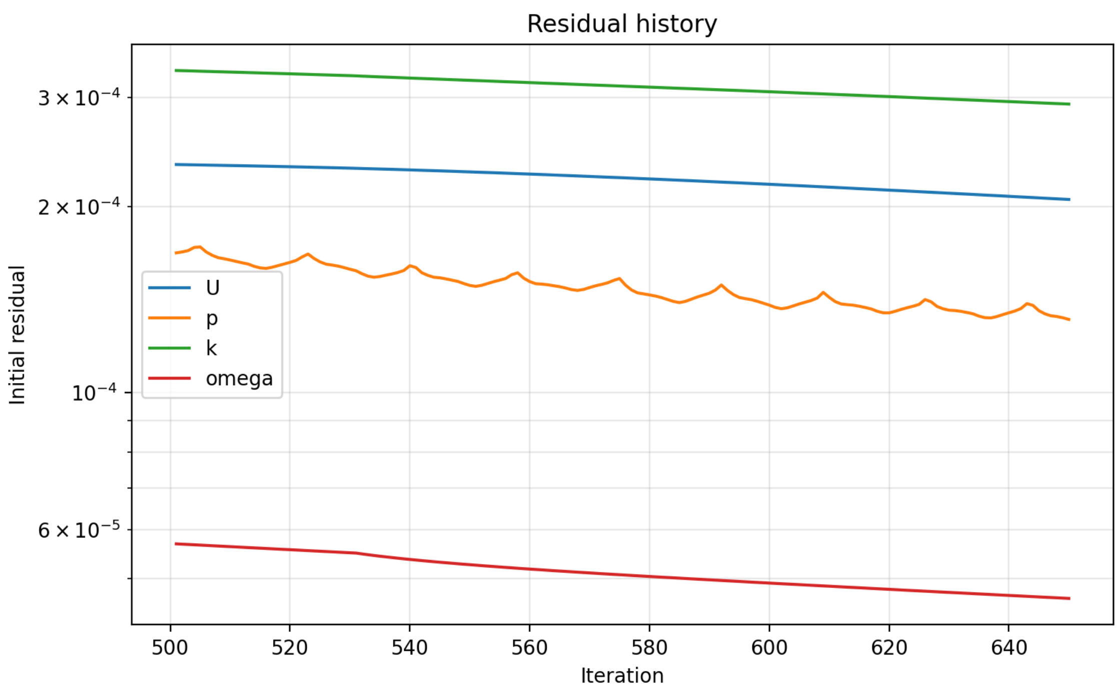

Figure 6: Residual history from the current best case, showing the deeper-convergence trajectory behind the pass-four winner.

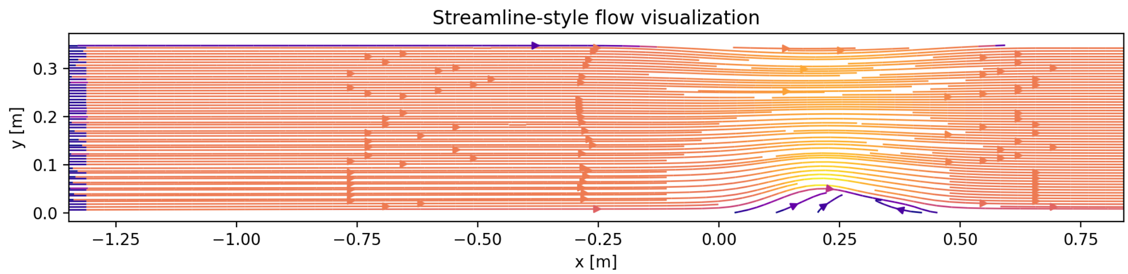

Figure 7: Streamline-style flow visualization from the promoted pass-four winner.

7 ParaView Outputs

The pass-four package also keeps the live `.foam`-based ParaView workflow. The screenshots below come from the case opened through `data/NASA_2DWMH/foam.foam` after the promoted best case was post-processed.

Figure 8: Pass-four ParaView velocity rendering from the `.foam` case.

Figure 9: Pass-four ParaView pressure rendering from the `.foam` case.

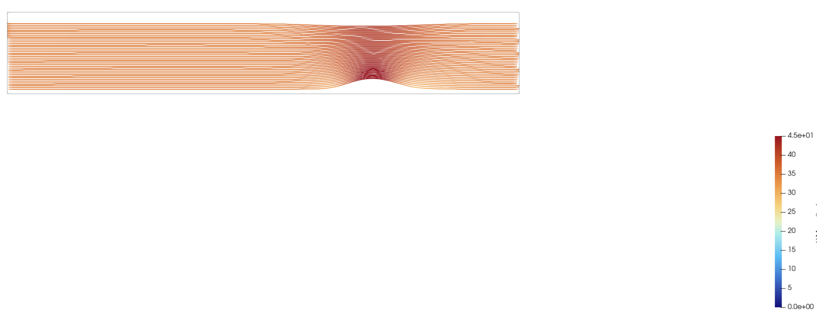

Figure 10: Pass-four ParaView streamline-style rendering from the `.foam` case.

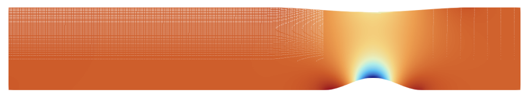

Figure 11: Pass-four ParaView mesh wireframe view from the `.foam` case.

Figure 12: Pass-four ParaView wall-focused rendering from the `.foam` case.

8 Discussion

The most important pass-four result is that the new experimental C_p/C_f comparison does not point to a cosmetic plotting issue. It points to a flow-field issue that still responds to additional convergence. The pass-four winner changed no geometry, no mesh topology, no turbulence closure, and no inlet model. It improved simply by continuing the already-best SST baseline from `500` to `650` iterations, and that improvement showed up simultaneously in:

- the local benchmark MAE;
- the wall-pressure comparison against the official C_p data;
- the wall-shear comparison against the official C_f data.

That makes the pass-four winner easier to defend than a more complicated configuration that changes many variables at once. It also confirms that the remaining error is still strongly tied to how the separated flow converges and recovers around the hump, not just to how the results are sampled.

The current mismatch is also clearer now than it was before the experimental tables were integrated. The solution still appears to:

- underpredict the front-half suction peak over parts of the hump;
- retain some wall-shear mismatch around the pre-separation and recovery regions;
- remain limited by the reconstructed inlet and geometry assumptions that are still approximate.

The direct comparison against the analytic-inlet screen is also useful. That case reduced the benchmark MAE relative to pass three, so the inflow boundary-layer issue is real. But it did not improve the official C_p and C_f agreement enough to beat the simpler continuation case. That is why pass four promotes the `650`-iteration SST continuation rather than the more complex inlet-profile variant.

9 Conclusion

Pass four delivered two concrete outcomes. First, it added a stricter and more honest comparison pipeline against the official experimental C_p and C_f tables using machine-readable NASA data only. Second, it improved the best local NASA hump MAE from `0.1801372055` to `0.1659136747` by extending the already-best SST case to `650` iterations.

The main lesson from pass four is that direct experimental wall-data comparison is valuable not because it creates a way to "aim" at a target curve, but because it helps diagnose which changes are genuinely improving the flow field. In this pass, the answer was conservative but clear: deeper convergence of the best existing baseline helped more than introducing a more elaborate inlet profile. That leaves a clean next step for any future pass: keep the strict experimental comparison pipeline, and only add new physical complexity if it beats the simpler converged baseline honestly.

10 Reproducibility

From the repository root, the pass-four workflow is:

```
python3 scripts/import_nasa_hump_experimental_data.py
bash scripts/run_nasa_hump_pass4.sh
bash scripts/render_nasa_hump_paraview.sh
bash scripts/build_report.sh
```

To rerun the focused pass-four experiment screen:

```
bash scripts/run_nasa_hump_pass4_experiments.sh
```

References